

A Analytical Study of Demands Patterns and Price Ambiguity in a Construction Material Retail Store

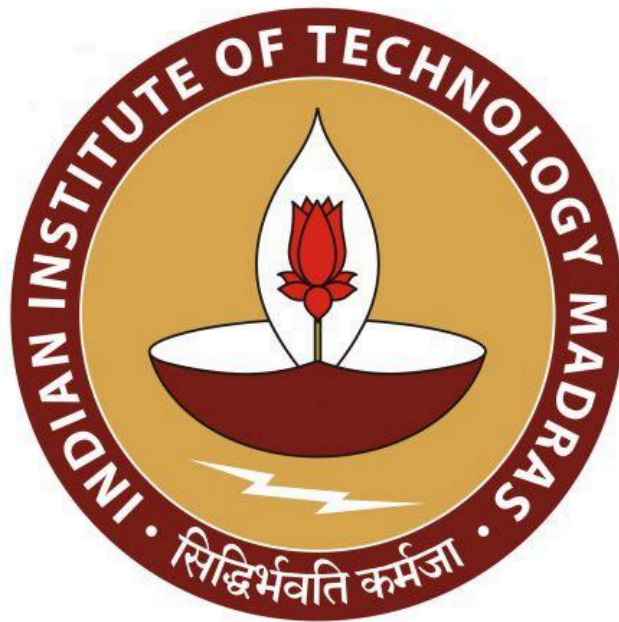
A Midterm Report for BDM Capstone Project

Submitted by

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Executive Summary

This study is based on Trishul Cement and Hardware Store, a local shop that sells construction materials like cement, TMT bars, stirrups, paints, and other related items. The store faces several practical challenges such as changing customer demand, fluctuating supplier prices, and difficulties in managing inventory properly. Demand for important products like cement and TMT bars is not the same throughout the year and is influenced by factors like monsoon season and construction activity. At the same time, changes in purchase prices make it difficult for the store owner to decide selling prices and maintain steady profits.

The analysis uses three main types of data: sales data, inventory data, and procurement data. The dataset includes columns such as product name, category, quantity sold, date, stock levels, purchase price, and selling price. Simple descriptive statistics like total sales, average demand, and revenue contribution of each product were used to understand the data. In addition, conversations with the store owner provided useful insights into real issues like stock shortages, supplier dependence, and pricing confusion.

Before analysis, the data was cleaned and properly organized. Monthly sales were studied using bar charts and line graphs to understand demand patterns. Profit margin analysis was used to compare selling and purchase prices, which showed that even though cement and TMT bars bring high revenue, their margins are not always stable due to price changes. Reorder point analysis was used to estimate when stock should be refilled, and it showed that stockouts happen during high-demand periods. Overall, the findings suggest that better planning of inventory and pricing can help improve the store's performance.

Proof of Data Originality

Name of Store: Trishul Cement and Hardware Store

Owner Name: Shreekant Kumar

Owner Contact No. : 9931886589



Image 1: Front View of the shop



Image 2: Storage Area

Supporting materials including video interaction with the store owner and images of transaction records are available at this link including video interaction with Owner and more Information about the shop

Link

:

https://drive.google.com/drive/folders/1OmVWjOZc9V7Hjg-kdLAm-Q20_gUHO7UB?usp=drive_link

(Kindly Copy this link and paste in a browser)

Also a transcript (in English) is provided in the link for better understanding

Letter from Organization

To Whom It May Concern

This is to certify that **Kishlay Kumar**(22f1001112) , is a bonafide student and is undertaking a **Business Data Management (BDM)** project as part of his academic curriculum.

This letter is issued to support the student in collecting and analyzing data from our business, Trishu Cement and Hardware Store, for his project work.

As part of the project, the student has interacted with us to understand our business operations, including sales, pricing, and inventory management. The data shared with him is based on actual business records and is being provided strictly for academic purposes.

We request that the information shared be used only for academic analysis and not for any commercial or external use.

I, **Kishlay Kumar**(22f1001112), confirm that the data collected will be used solely for academic purposes and will not be shared with any third party.

For Trishul Cement & Hardware Store

Shreekanth Kumar

Proprietor

Signature of Shreekanth kumar

Owner , Trishul Cement and Hardware Store

Kishlay Kumar

Signature of Kishlay

Student , IITM BS

Image 3 : Authorisation Letter from Organization

Metadata



Image 4: Overview of Data Organization

Raw Sales Data: Sales data refers to daily transaction records collected from the hardware shop.

Content: The data was collected from handwritten bills and digitized using Google Sheets. A total of **721 bills** were recorded, forming a dataset of **721 rows and 9 columns**. It includes variables such as date, product, brand, quantity, unit, cost price, selling price, revenue and profit.

Justification: This data helps analyze demand patterns, brand performance, and profitability, supporting business decisions.

Cleaned Data: The raw sales data was cleaned and consolidated into a date-wise, product-level format to ensure consistency and usability.

Content: The dataset includes key variables such as date and month, which help track how demand changes over time and capture seasonal patterns. Product-wise quantities for categories like cement, TMT bars, stirrups, paint, and others provide detailed insights into daily sales behavior, allowing identification of fast- and slow-moving items. The category-wise structure further organizes the data for easier comparison across product groups, while the daily total reflects overall demand intensity and highlights peak and low sales periods. Together, these variables enable a clear understanding of demand fluctuations, indicate shifts in customer preference possibly influenced by pricing differences, and support effective inventory management through data-driven decision-making.

Justification: These variables help in performing reorder point (ROP) analysis by providing clear insights into demand patterns and variability. By organizing data in a structured, time-based format, it becomes easier to estimate average demand and fluctuations, allowing more accurate and practical inventory decisions.

Inventory Data: Inventory data refers to stock-level records of products in the shop.

Content: The data was derived from sales data and maintained on a weekly basis. It includes opening stock, closing stock, used to estimate stock usage and support reorder analysis.

Justification: This data helps monitor stock levels, identify shortages, and determine reorder requirements for efficient inventory management.

Purchase Data: Purchase data refers to records of materials purchased by the shop.

Content: The dataset contains weekly purchase details for products such as cement, TMT bars, stirrups, and paints. It includes variables representing quantities purchased for different brands and product types over specific time periods.

Justification: This data helps analyze purchasing patterns, supplier demand, and stock replenishment. It also supports reorder analysis by linking purchases with inventory and sales data.

Calculations:

Content:

The calculations sheet contains tabulated results such as descriptive statistics (mean, median, mode, standard deviation, etc.), along with computed profit and profit margin values. It also includes summary tables for inventory movement and reorder analysis.

Justification:

These calculations provide a clear numerical summary of the data, helping to identify demand patterns, pricing variations, and inventory-related issues.

Visuals:

Content:

The visuals sheet includes charts showing monthly revenue and profit trends, product-wise sales comparisons, and profit analysis.

Justification:

These charts make it easier to understand trends and patterns, supporting better interpretation and decision-making.

Dataset Link:

https://docs.google.com/spreadsheets/d/1OmjL2IAWmr2Rfo-kMSmR1bxhCRpGWaFh/edit?usp=drive_link&ouid=103673603373318705137&rtpof=true&sd=true(copy the link and paste it in a browser)

Descriptive Statistics

Product	Mean	Median	Mode	Standard Deviation	Min	Max	Range	Sum	Count (Sale Days)
Nuvoco	6.15	5	2	15.41	1	100	100	1131	74
UltraTech	5.82	10	2	15.55	1	100	100	1071	56
JSW	6.21	8	1	15.65	1	100	100	1142	73
Concreto	1.84	16	4	7.15	2	50	50	338	19
TMT_8mm	3	46	46	13.26	46	92	92	552	10
TMT_10mm	35.16	71	71	65.23	71	355	355	6470	55
TMT_12mm	42.4	52	51	77.58	50	561	561	7801	75
Stirrup_Square	28.48	100	50	60.12	15	300	300	5240	52
Stirrup_Circular	36.03	87.5	25	75.04	10	400	400	6629	62
Stirrup_Rect	49.55	100	200	84.13	10	450	450	9117	73
Paint	3.39	10	5	6.44	2	30	30	624	59
Primer	3.07	6	10	5.08	1	30	30	564	73
Binding_Wire	0.73	10	10	2.64	5	20	20	135	16
Tarpaulin_Cover	0.26	4	2	1.07	2	8	8	47	12

Image 5 : Descriptive statistics

Cement: Cement is sold in standard 50 kg bags across all brands, ensuring uniformity in quantity. Demand across brands is quite similar, with mean values ranging from about 5 to 6. Three brands show similar levels of variability, while Concreto has a relatively lower standard deviation, indicating slightly more stable demand compared to the others. However, overall variability remains high, with a wide gap between minimum and maximum values and peaks close to 100 units, showing that sales are not consistent.

This variation is influenced not only by construction activity , seasonal changes(like monsoon and winter) and festivals, but also by price ambiguity across brands. When prices differ or are unclear, customers tend to switch between options, leading to uneven sales. As a result, some brands experience sudden demand spikes while others remain unsold, making inventory management more challenging.

TMT Bars: TMT bars are typically sold in fixed bundle sizes—8mm comes in bundles of 10 bars weighing around 45–47 kg, 10mm in 7 bars weighing 70–72 kg, and 12mm in 5 bars weighing 50–52 kg. This pattern is clearly reflected in the data. For instance, 8mm shows relatively low and steady demand, with a mean of 3 and a standard deviation of 13.26, indicating smaller and more regular purchases. In contrast, 10mm has a mean of 35.16 and a standard deviation of 65.23, while 12mm has a mean of 42.4 and a standard deviation of 77.58, both showing much higher variability. The maximum values reaching 355 kg for 10mm and 561 kg for 12mm indicate frequent bulk purchases, likely driven by construction projects. The wide range between minimum and maximum values further confirms that demand is irregular and influenced by occasional large orders.

Stirrups: Stirrups show moderate to high demand across types, with rectangular stirrups having the highest mean of 49.55, followed by circular at 36.03 and square at 28.48. All three types exhibit high variability, with standard deviation values of 84.13 for rectangle stirrup , 75.04 for circular, and 60.12 for square. The wide range between minimum and maximum values, with peaks reaching above 400 units, indicates that demand is not consistent.

This pattern reflects the construction based nature of stirrup usage, where specific shapes are required depending on construction needs. As a result, demand occurs in bursts, leading to sudden spikes in certain types while others see lower movement. This also explains the stockouts and backordering observed in circular and rectangular stirrups, which are **highlighted in red in the inventory data**, making inventory planning more challenging.

Others: Primer (mean: 3.07, SD: 5.08), Paint (mean: 3.39, SD: 6.44), Binding Wire (mean: 0.73, SD: 2.64), and Cover Blocks (mean: 0.26, SD: 1.07) exhibit relatively low average demand with comparatively lower variability than core construction materials. This indicates more stable and consistent consumption patterns.

These products are primarily supporting materials, with demand closely tied to ongoing construction activities rather than independent bulk purchasing. Unlike cement or TMT bars, they are typically purchased in smaller quantities and at more regular intervals, resulting in fewer extreme fluctuations.

Detailed Explanation of Analysis Process

1. Data Cleaning and Preprocessing

Handwritten bills were digitized using Google Sheets to enable structured analysis. During the cleaning process, incorrect and inconsistent entries were identified and corrected based on domain knowledge. For instance, the standard weight of a bundle of TMT 8mm bars (10 bars) ranges between 45–47 kg; any values outside this range were flagged and corrected. Similarly, cement bags across all brands have a fixed weight of 50 kg, and any deviations (e.g., 55 kg) were treated as data entry errors and resolved.

Justification: Data cleaning was performed to ensure accurate recording of transactions, thereby maintaining correct opening and closing stock levels. This is essential for effective inventory management and for preventing issues such as stockouts and excess inventory.

2. Descriptive Statistics

Key metrics such as mean, range, and standard deviation were calculated for all product categories to analyze demand patterns. Cement showed similar mean demand with high variability, TMT bars (especially 10mm and 12mm) reflected bulk-driven fluctuations, stirrups showed shape-based variation, and other items like primer and binding wire had stable demand.

Justification: This analysis helps identify demand variability and patterns across products, highlighting issues such as demand fluctuation, price ambiguity, and inventory challenges. These insights support better stock planning by reducing stockouts and excess inventory.

3. Reorder Analysis

Reorder levels were estimated using mean demand, standard deviation, and a lead time of 5 days from the cleaned dataset. These values indicate the stock level at which a new order should be

placed to avoid both stockouts and excess inventory. Cement products show moderate levels such as Nuvoco 56.17 and UltraTech 54.75, suggesting balanced demand. In contrast, TMT 10mm and 12mm show much higher levels of 283.42 and 340.00, while stirrup rectangular and circular reach 386.56 and 303.96, reflecting bulk and irregular purchasing patterns.

Formula Used: $ROP = (\text{Mean} \times \text{Lead Time}) + \text{Safety Stock}$

Justification: This approach helps the store owner reorder at the right time—before stock runs out and without overstocking. Higher levels for TMT bars and stirrups explain the backordering and stockouts already observed, while controlled levels for cement reduce unnecessary excess stock despite demand fluctuations. Overall, it provides a practical way to manage both overstocking and backordering using actual demand patterns.

4. Profit Margin Analysis

The profit margin analysis shows that the shop is keeping higher margins than typical market levels to maximize profit. Generally, cement has a base margin of 12–15%, TMT bars around 4–5%, and stirrups 10–20%. However, in this dataset, cement margins go up to 19.38% (UltraTech) compared to 13.13% (Concreto). TMT bars range between 5.45%–6.78%, slightly above normal levels. The biggest difference is seen in stirrups, where Square and Rectangle reach 33.33%, far above the expected range, while Circle is only 9.08%. Compared to nearby shops offering lower prices, this indicates that the store is pricing products above standard market levels.

Justification:

Profit margin analysis was used because it clearly shows how much profit is earned on each product, making it easier to understand pricing decisions. It also allows comparison across different products and with nearby shop prices. When margins are higher than normal, customers may shift to cheaper alternatives, leading to potential loss of sales. This makes it difficult for customers to judge a fair price, resulting in price ambiguity and affecting long-term demand.

Results and Findings

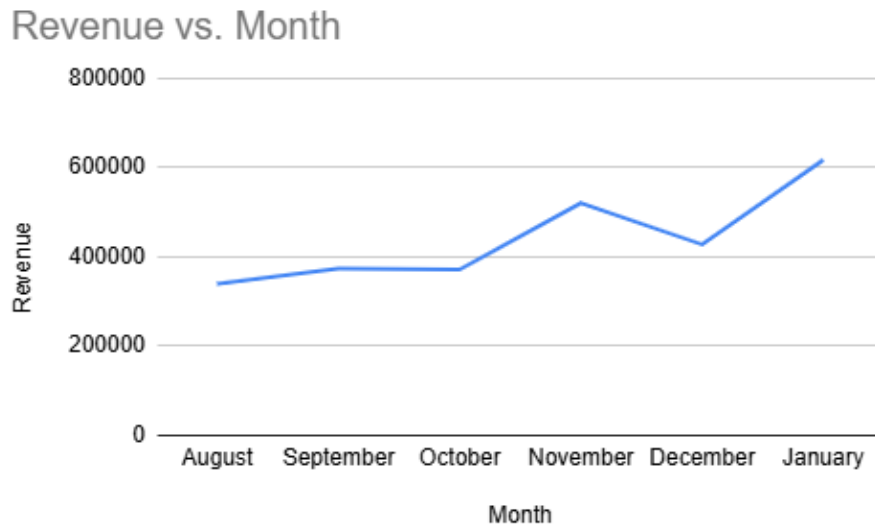


Image 6: Revenue and Profit

The graph shows the revenue earned by the store from August 2025 to January 2026. It can be observed that revenue during August and September is relatively lower, which may be due to the monsoon season when construction activities slow down. As the monsoon ends, revenue gradually increases from October onwards, indicating a rise in demand during the pre-winter construction period. The trend continues upward, with November and then decline with December due to the Winter Season. January records the highest revenue and profit, suggesting strong business activity during this period. Overall, the graph highlights a clear seasonal pattern where business is slower during monsoon and improves steadily towards the winter season.

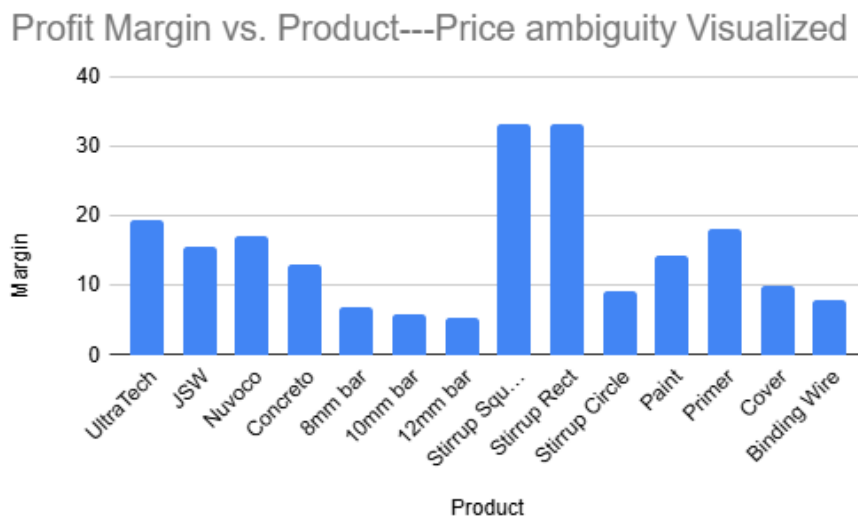


Image 7 : Price ambiguity Visualization

The graph shows the variation in profit margins across different products sold by the store. It can be observed that the profit margin is not consistent across product categories. Some products, such as certain types of cement and stirrups, have relatively higher margins, while others have lower margins.

This inconsistency indicates that the store does not follow a clear or standardized pricing strategy across all products. In several cases, the margins appear higher than what is typically expected in the local market. This suggests that pricing decisions may be influenced by fluctuating procurement costs or a lack of structured pricing policies.

Such variation in margins highlights the issue of price ambiguity, where there is no fixed relationship between cost and selling price. As a result, customers may experience inconsistent pricing, and the store may face difficulty in maintaining stable profits.

Overall, the analysis shows the need for a more consistent pricing approach, where profit margins are better aligned across products while still considering market conditions and competition.

